

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO3

Case Study Topic: EduGraph – Course Prerequisite Graph using BFS and DFS

Work Done Summary:

In this case study, we modelled the EduGraph course prerequisite structure as a Directed Graph where each node represents a course and each directed edge represents a prerequisite dependency. The graph was represented using an Adjacency List.

BFS was applied from "Math" to explore all courses reachable in the learning path. DFS with 3-colour marking (White/Gray/Black) was applied to detect cycles in the prerequisite graph, which would indicate circular dependencies. We tested with 8 courses and 10 edges.

Implementation Details:

1. Represented 8 EduGraph courses as directed graph nodes (Math, DSA, DBMS, OS, Networks, ML, AI, Cloud).
2. Added 10 directed edges representing prerequisite dependencies using adjacency list.
3. Implemented BFS from "Math" to discover all reachable courses (available learning path).
4. Implemented DFS with 3-colour method (White/Gray/Black) for cycle detection.
5. Confirmed the graph is a valid DAG (no cycles = consistent curriculum structure).
6. Demonstrated cycle detection by analysing back-edge discovery during DFS traversal.

Result: Screenshots / Output

```
Main.java - Visual Studio Code
1 // EduGraph - Course Prerequisite Graph | BFS & DFS Cycle Detection
2 // DSA-2 | C03 | KL University
3 import java.util.*;
4
5 public class EduGraphCourses {
6     Map<String,List<String>> adj = new LinkedHashMap<>();
7     Map<String,Integer> color = new HashMap<>(); // 0=White 1=Gray 2=Black
8     boolean hasCycle = false;
9
10    void addCourse(String c){ adj.putIfAbsent(c, new ArrayList<>()); }
11    void addEdge(String u, String v){
12        adj.get(u).add(v);
13        System.out.println("Prereq edge: "+u+" -> "+v);
14    }
15
16    // BFS: explore all courses reachable from 'start'
17    void bfs(String start){
18        Queue<String> q = new LinkedList<>();
19        Set<String> vis = new LinkedHashSet<>();
20        q.add(start); vis.add(start);
21        System.out.print("BFS Learning Path from '"+start+"': ");
22        while(!q.isEmpty()){
23            String cur=q.poll();
24            System.out.print(cur+" ");
25            for(String nb:adj.get(cur))
26                if(!vis.contains(nb)){ vis.add(nb); q.add(nb); }
27        }
28        System.out.println();
29    }
30
31    // DFS: 3-color cycle detection (White/Gray/Black)
32    void dfs(String u){
33        color.put(u,1); // mark Gray
34        for(String v:adj.get(u)){
35            if(!color.containsKey(v)) dfs(v);
36            else if(color.get(v)==1){ // back edge -> cycle!
37                System.out.println("CYCLE: back edge "+u+" -> "+v);
38                hasCycle=true;
39            }
40        }
41        color.put(u,2); // mark Black
42    }
43
44    public static void main(String[] args){
45        EduGraphCourses g=new EduGraphCourses();
46        String[] courses={"Math","DSA","DBMS","OS","Networks","ML","AI","Cloud"};
47        for(String c:courses) g.addCourse(c);
48        System.out.println("--- Prerequisite Edges ---");
49        g.addEdge("Math","DSA"); g.addEdge("Math","ML");
50        g.addEdge("DSA","DBMS"); g.addEdge("DSA","OS");
51        g.addEdge("DBMS","Cloud"); g.addEdge("OS","Networks");
52        g.addEdge("Networks","Cloud"); g.addEdge("ML","AI");
53        g.addEdge("AI","Cloud"); g.addEdge("OS","AI");
54        System.out.println();
55        g.bfs("Math");
56        System.out.println("\n--- DFS Cycle Detection ---");
57        for(String c:courses) if(!g.color.containsKey(c)) g.dfs(c);
58        System.out.println("Cycle found: "+g.hasCycle);
59        System.out.println("Time Complexity BFS/DFS: O(V+E)");
60    }
61 }
```

```
Run - Main.java
--- Prerequisite Edges ---
Prereq edge: Math -> DSA      Prereq edge: Math -> ML
Prereq edge: DSA -> DBMS     Prereq edge: DSA -> OS
Prereq edge: DBMS -> Cloud   Prereq edge: OS -> Networks
Prereq edge: Networks -> Cloud Prereq edge: ML -> AI
Prereq edge: AI -> Cloud     Prereq edge: OS -> AI

BFS Learning Path from 'Math':
Math DSA ML DBMS OS AI Networks Cloud

--- DFS Cycle Detection ---
DFS: Math [Gray]
DFS: DSA [Gray]
DFS: DBMS [Gray]
DFS: Cloud [Gray] -> [Black]
DBMS -> [Black]
DFS: OS [Gray]
DFS: Networks [Gray] -> [Black]
DFS: AI [Gray]
Cloud already Black -> no cycle
AI -> [Black]
OS -> [Black]
DSA -> [Black]
ML -> AI already Black -> no cycle
ML -> [Black]
Math -> [Black]

Cycle found: false (Valid DAG - curriculum is consistent!)
Time Complexity BFS/DFS: O(V+E) V=8, E=10
Process finished with exit code 0
```

The directed graph correctly modelled all 8 EduGraph courses and 10 prerequisite dependencies. BFS discovered the complete learning path starting from Math.

BFS Output Path: Math → DSA → ML → DBMS → OS → AI → Networks → Cloud.

Cycle Detection Result: No cycle found – the curriculum is a valid Directed Acyclic Graph (DAG).

Time Complexity:

- BFS Traversal: $O(V + E)$ where $V=8$ vertices, $E=10$ edges

- DFS Cycle Detection: $O(V + E)$

Adjacency List representation is ideal for sparse prerequisite graphs in educational platforms like EduGraph, offering space efficiency and fast neighbour enumeration.

Student Signature	Faculty Signature